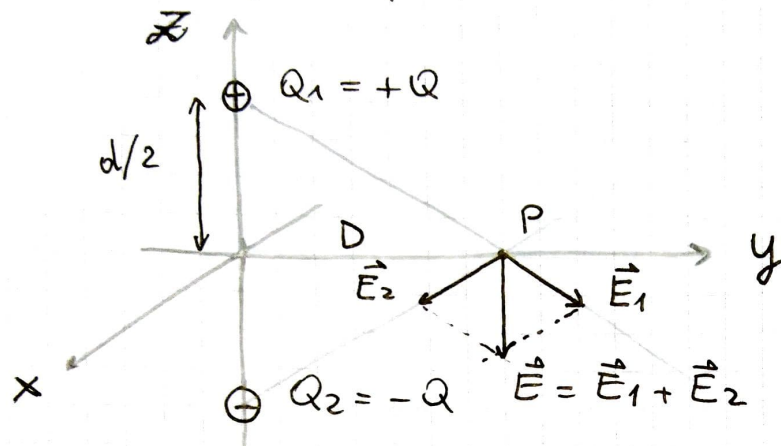


Esercizio 3.

(Campo elettrostatico / principio di sovrapposizione)
cariche puntiformi



2 cariche elettriche puntiformi, pari modulo ma segno opposto, si trovano a distanza $(d/2)$ dall'origine lungo l'asse z , simmetricamente rispetto all'origine. Calcolare il campo elettrico da esse generato nel punto P dell'asse delle y posto a distanza D dall'origine.

$$\vec{E}_0(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{Q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i)$$

$$E_{0j}(x, y, z) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{Q_i (j - j_i)}{[(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2]^{3/2}}$$

$$j = \{x, y, z\}$$

$$P \mapsto \vec{r} \equiv (x, y, z) = (0, D, 0)$$

$$Q_1 \mapsto \vec{r}_1 \equiv (x_1, y_1, z_1) = (0, 0, d/2)$$

$$Q_2 \mapsto \vec{r}_2 \equiv (x_2, y_2, z_2) = (0, 0, -d/2)$$

$$|\vec{r} - \vec{r}_1| = \sqrt{(D-0)^2 + (0-d/2)^2} = \sqrt{D^2 + \left(\frac{d}{2}\right)^2}$$

$$|\vec{r} - \vec{r}_2| = \sqrt{D^2 + \left(\frac{d}{2}\right)^2} \quad \Delta = \sqrt{D^2 + (d/2)^2}$$

Calcoliamo le componenti della risultante:

$$E_{0x} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q \cdot (0-0)}{\Delta^3} - \frac{Q \cdot (0-0)}{\Delta^3} \right] = 0$$

$$E_{0y} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q(D-0)}{\Delta^3} - \frac{Q(D-0)}{\Delta^3} \right] = 0$$

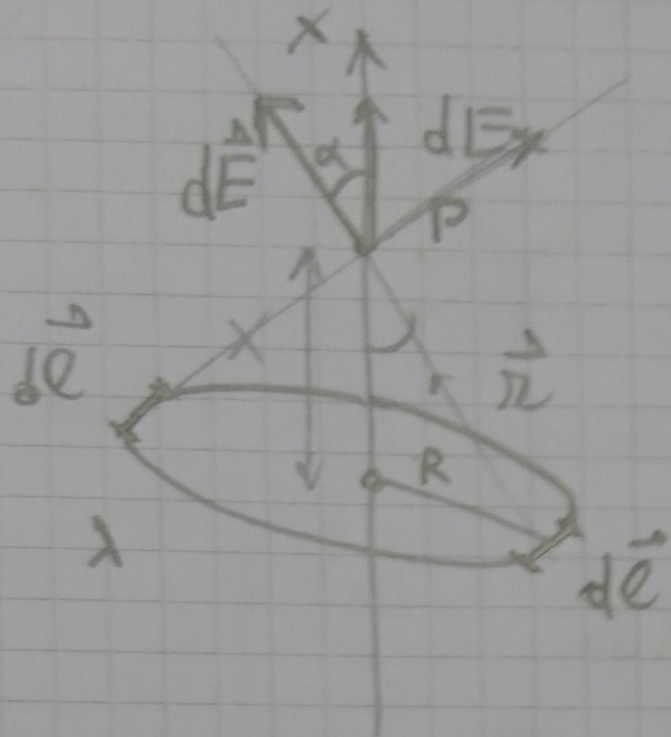
$$E_{0z} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q(0-d/2)}{\Delta^3} - \frac{Q(0+d/2)}{\Delta^3} \right] = \frac{1}{4\pi\epsilon_0} \cdot \frac{-Qd}{\Delta^3}$$

$$\vec{p} = Q \cdot \vec{d} \quad \text{momento di dipolo elettrico}$$

$$\vec{E}_0 = \frac{-\vec{p}}{4\pi\epsilon_0 \left[D^2 + \left(\frac{d}{2}\right)^2 \right]^{3/2}}$$

Esercizio 5.

Calcolare il campo elettrico generato dall'asse da una distribuzione uniforme di carica distribuita su una spira circolare filiforme.

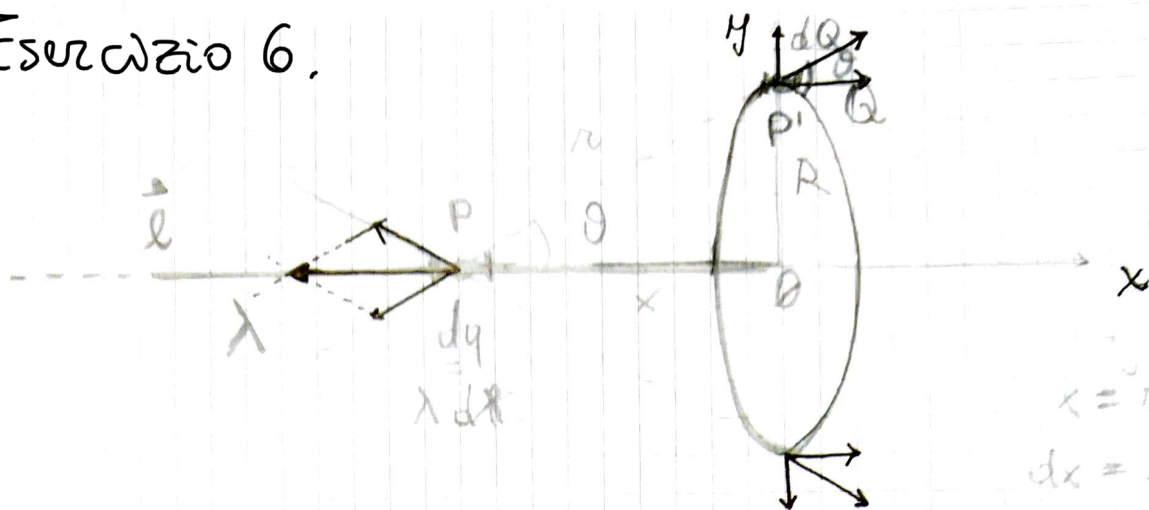


$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dl}{r^2} \hat{r}$$

$$dE_x = dE \cdot \cos\alpha = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dl}{r^2} \cos\alpha$$

$$\begin{aligned} |\vec{E}| &= E_x = \int dE_x = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda \cos\alpha}{r^2} \int dl = \frac{2\pi R}{2\pi\epsilon_0} \cdot \frac{\lambda \cos\alpha}{r^2} = \\ &= \frac{\lambda R}{2\epsilon_0} \cdot \frac{x}{r^3} = \frac{\lambda R}{2\epsilon_0} \cdot \frac{x}{(x^2 + R^2)^{3/2}} \xrightarrow{x \gg R} \frac{\overbrace{2\pi R \lambda}^Q}{4\pi\epsilon_0} \cdot \frac{1}{x^2} \end{aligned}$$

Esercizio 6.



$$x = r \cos \theta = R \cos \theta$$

$$dx = -R \sin \theta d\theta$$

$$R = \frac{R}{\sin \theta}$$

$$E_a^P = \frac{Q}{4\pi\epsilon_0} \cdot \frac{x}{(x^2 + R^2)^{3/2}}$$

$$dF_a = dq \cdot E_a = E_a \cdot \lambda dx$$

$$F_a = \int_0^\infty E_a \lambda dx = \frac{Q \lambda}{4\pi\epsilon_0} \int_0^\infty \frac{x dx}{(x^2 + R^2)^{3/2}} =$$

$$= \frac{Q \lambda}{4\pi\epsilon_0} \left[-\frac{1}{(x^2 + R^2)^{1/2}} \right]_0^\infty = \frac{Q \lambda}{4\pi\epsilon_0 R}$$

$$dE_{filo}^P = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{r^2} \cdot \cos \theta$$

$$E_{filo} = + \int_0^{\pi/2} \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda dx}{r^2} \cdot \cos \theta d\theta = -\frac{\lambda}{4\pi\epsilon_0} \int_0^{\pi/2} \frac{\sin^2 \theta R}{R^2 \sin \theta} \cos \theta d\theta$$

$$= -\frac{\lambda}{4\pi\epsilon_0 R} [\sin \theta]_0^{\pi/2} = -\frac{\lambda}{4\pi\epsilon_0 R}$$

$$F_{filo} = Q \cdot E_{filo} = -\frac{\lambda Q}{4\pi\epsilon_0 R}$$

$$F_{filo} = -F_a$$